

On the endoscopic classification of automorphic representations of quasi-split classical groups: co-tempered A-packets

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§. Automorphic forms

G conn. reductive group / $\mathbb{Q}$

$$L^2 \left(\underbrace{\Gamma \backslash G(\mathbb{R})}_{\text{finite volume}} \right) \hookrightarrow G(\mathbb{R})$$

$$\Gamma = G(\mathbb{Z})$$

$\downarrow$
cuspidal
mod. forms

$$G(\mathbb{Z}) \backslash G(\mathbb{R}) \xrightarrow{\text{strong approx}}$$

$$\cup$$

$$G(\mathbb{Q}) \backslash G(\mathbb{A}) / G(\hat{\mathbb{Z}})$$

$$\mathbb{A} = \mathbb{R} \times \prod_{p < \infty} \mathbb{Q}_p$$

$$\hat{\mathbb{Z}} = \prod_{p < \infty} \mathbb{Z}_p$$

$$L^2(G)$$

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$$L^2 \left(\underbrace{G(\mathbb{Q}) \backslash G(\mathbb{A})}_{\text{finite volume}} \right) \hookrightarrow G(\mathbb{A}) = G(\mathbb{R}) \times \prod_{p < \infty} G(\mathbb{Q}_p)$$

finite volume

$$L^2(G) := \underbrace{L^2_{\text{disc}}(G)} \oplus \underbrace{L^2_{\text{cts}}(G)}$$

$$\begin{aligned} \oplus \pi^{m(\pi)} &\xrightarrow{\text{Eisenstein series}} \int^\oplus \pi d\mu(\pi) \\ &\uparrow \\ &\oplus_{\substack{M \subset G \\ \text{Levi}}} I_P^G(L^2_{\text{disc}}(M)) \end{aligned}$$

§. Langlands' conjectures

Fact. Any irrep π of $G(\mathbb{A})$ decomposes as $\bigotimes_{p \leq \infty} \pi_p$, π_p irrep of $G(\mathbb{Q}_p)$.

Conj. (reciprocity) $\text{Irr}(\text{const}(L^2_{\text{disc}}(G))) \xleftrightarrow{\pi \longleftrightarrow \psi} \text{Irr}(\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}), \hat{G})$

Langlands

s.t. $L(s, \pi, \tau)$

$$= L(s, \tau \circ \psi)$$

Artin

$$\pi \mapsto \pi_p \downarrow$$

$$\text{Irr}(G(\mathbb{Q}_p)) \longleftrightarrow \text{Rep}(\text{Gal}(\bar{\mathbb{Q}}_p/\mathbb{Q}_p), \hat{G})$$

$$\psi \mapsto \psi|_{D_p}$$

$$\forall \tau: \hat{G} \rightarrow GL_N.$$

Conj. (functoriality) Given $\zeta: \hat{H} \rightarrow \hat{G}$, we should have

$$\zeta_*: \text{Irr}(\text{const}(L^2(H))) \rightarrow \text{Irr}(\text{const}(L^2(G)))$$

$$\text{s.t. } L(s, \zeta_*(\pi), \tau) = L(s, \pi, \tau \circ \zeta)$$

Applications

a) Shimura varieties aut forms $\longleftrightarrow$ Galois reps

motives

b) Abelian surfaces (Boxer - Gee - Pilloni)

c) Even unimodular lattices: $L \mid \mathbb{Z}$, $\eta: L \otimes L \rightarrow \mathbb{Z}$

$$L \rightsquigarrow SO(L) = G \text{ smooth reductive } / \mathbb{Z} \quad \eta(x, x) \in 2\mathbb{Z}$$

$$\{L^1 \mid \text{rk}, \text{disc} = L\} \xleftrightarrow{1:1} G(\mathbb{Q}) \backslash G(\mathbb{A}) / G(\hat{\mathbb{Z}})$$

finite set

(Chenavier - Lannes)

$$L^2(G(\mathbb{Q}) \backslash G(\mathbb{A}) / G(\hat{\mathbb{Z}}))$$

d) Bloch - Kato

Arthur's result:

$$\text{Rep}(\text{Gal}_F, \hat{G}) \leftrightarrow \mathbb{I}(G) \stackrel{\text{def}}{=} \hat{G} \rtimes \text{Gal}_F$$

$$\left\{ \psi: L_F \times \text{SL}_2^A \longrightarrow L_G \right\} / \hat{G}\text{-conj.}$$

/ bounded

$$W_{\mathbb{R}}, W_{\mathbb{Q}_p} \times \text{SL}_2^D, \text{ etc.}$$

$$\text{Irr}(\text{Gal}_F, \hat{G}) \mapsto \mathbb{I}_2(G) = \left\{ \overbrace{\text{Cent}(\psi, \hat{G})}^{S_\psi} / \mathbb{Z}(\hat{G}) \text{ finite} \right\}$$

$$\text{Unit}(\text{Gal}_F, \hat{G}) \mapsto \mathbb{I}_{\text{temp}}(G) = \left\{ \psi \mid \text{SL}_2^A = 1 \right\}$$

Th. (Arthur, conditional) $G = \text{Sp}, \text{SO}, \text{U}$ (Mek)

$$1) L_{\text{disc}}^2(G) = \bigoplus_{\psi \in \mathbb{I}_2(G)} L_\psi^2(G)$$

$$2) \text{Irr}_{\text{Aut}}(G(\mathbb{A}_p)) = \bigcup_{\psi_p \in \mathbb{I}(G_{\mathbb{A}_p})} \pi_\psi$$

$$\text{Irr}_{\text{temp}}(G(\mathbb{A}_p)) = \bigsqcup_{\psi_p \in \mathbb{I}_{\text{temp}}(G_{\mathbb{A}_p})} \pi_\psi$$

$$3) \pi_p \mapsto \rho \pi_p$$

$$\pi_{\psi_p} \longrightarrow \text{Irr}(\text{BS}_{\psi_p})$$

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inj. if ψ temp

bij. if $p < \infty$

$$4) L_\psi^2(G) = \bigoplus_{\substack{\pi \in \text{Irr}(G(\mathbb{A})) \\ \pi_p \in \pi_{\psi_p}}} \pi \otimes m(\psi, \pi)$$

$$m(\psi, \pi) = \text{mult}(\Sigma_\psi, \bigotimes_{p \leq \infty} \rho \pi_p)$$

$$\Sigma_\psi: S_\psi \longrightarrow \{\pm 1\}$$

$$\Sigma_\psi \equiv 1 \text{ } \psi \text{ temp.}$$

§ Method of proof

$$f \in C_c^\infty(G(A))$$

$$TF^G(f) : \sum_Y O_Y(f) + \text{supp} = \sum_\pi \text{tr} \pi(f) + \text{supp}$$

$$\begin{array}{ccc} \int f(grg^{-1}) & \xleftrightarrow{WGT} & \text{tr} \left(\int f(x) \pi(x) \right) \\ \uparrow \text{GT} & & \uparrow \text{ECR} \end{array} \quad \begin{array}{c} \uparrow \\ \text{LIR} \end{array}$$

$$TF^{G'}(f)$$

$$GT \Leftarrow FL : Ng0$$

$$WGT \Leftarrow WFL : Choudhury - Lannan G split$$

$$\text{Conner Haleck - Dube } G \text{ q-split}$$

$$\text{ECR } \psi, s \in S_\psi \rightsquigarrow \hat{G}' = \text{Cent}(s, \hat{G})^\circ, G', \psi'$$

$$\sum_{\pi \in \Pi_\psi} \text{tr} P_\pi(s, S_\psi) \cdot \text{tr} \pi(f) = \sum_{\pi' \in \Pi_{\psi'}} \text{tr} P_{\pi'}(S_{\psi'}) \text{tr} \pi'(f')$$

$$\text{LIR} : \psi^M : \mathbb{F}(M) \rightarrow \mathbb{F}(G)$$

$$\sum_{\pi^M \in \Pi_{\psi^M}} \text{tr} (R_p(s) \cdot I_p^G(\pi^M, f))$$

ψ tempered: Robust theory over $\mathbb{R}$ (Langlands + Shalika)

ψ non-tempered: ??

§ Our results $G/\mathcal{O}_p$ $\mathcal{H} = C_c^\infty(G) \rightsquigarrow \pi$ rep of G

Aubert duality

$$D: D(\text{Rep}(G)) \longrightarrow D(\text{Rep}(G))$$

$$\pi \longmapsto \text{RHom}_{\mathcal{H}}(\pi, \mathcal{H})$$

$$D(\pi) = \beta(\pi) \cdot \hat{\pi}[d]$$

$$\pi \text{ irred.} \Rightarrow \hat{\pi} \text{ irred.}$$

$$\pi \text{ temp.} \not\Rightarrow \hat{\pi} \text{ temp.}$$

$$\psi: W_{\text{clp}} \times SL_2^D \times SL_2^A \longrightarrow L_G$$

$$\hat{\psi}(\omega, a, b) = \psi(\omega, b, a)$$

$$\text{Conj. } \pi_{\hat{\psi}} = \{ \hat{\pi} : \pi \in \pi_{\psi} \}$$

Th. (AGIKMS) : Assume Arthur's thm for tempered parameters.

Let ψ be ω -tempered, i.e. $\hat{\psi}$ is tempered

$$1) \text{ Define } \pi_{\psi} = \{ \hat{\pi} : \pi \in \pi_{\hat{\psi}} \}$$

There exists $\pi_{\psi} \longrightarrow \text{Irr}(S_{\psi})$ s.t. ECR holds.

2) If $\psi^M \in \mathbb{F}(M) \rightarrow \mathbb{F}(G) \ni \psi$, then LIR holds.

$$\begin{array}{ccc} \text{Proofs. } 1) & \pi_{\psi} & \xleftarrow{\pi \leftrightarrow \hat{\pi}} \pi_{\hat{\psi}} \\ & \uparrow & \times \quad \downarrow 1-1 \\ & \text{Irr}(S_{\psi}) & = \text{Irr}(S_{\hat{\psi}}) \end{array}$$

$$\text{tr } p_{\pi}(s) := \text{tr } p_{\hat{\pi}}(s).$$

$$\beta(\psi) \beta(\psi^{s_2+1}) \beta(\psi^{s_2-1})$$

$$\begin{array}{c} S_{\psi} \\ \downarrow \\ \text{tr } p_{\pi}(S_{\psi}) = \beta(\pi) \cdot \beta(\psi) \\ / \\ \psi(1, [-1, 1]) \end{array}$$

